

Yuru Sun

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SUMMARY

Quantitative researcher with experience developing, validating and applying statistical and predictive models to complex financial data. Expertise includes nonlinear state-space modelling, Bayesian statistics, probabilistic forecasting, simulation and out-of-sample model evaluation. Proficient in R and MATLAB, with experience in Python and SQL, and in building reproducible workflows for large, complex and imperfect datasets across equity, fixed-income and derivatives markets.

SELECTED QUANTITATIVE RESEARCH EXPERIENCE

Modelling Financial Price Jumps and Tail Risk Forecasting 2022 - 2025

- Developed and estimated nonlinear probabilistic models for rare financial events using Bayesian mixture and Markov-switching methods, allowing latent risk states and event distributions to be learned from data.
- Outperformed conventional parametric benchmarks in out-of-sample density and tail forecasting across four financial indices, with particularly strong performance during the Global Financial Crisis and COVID-19.
- Conducted simulation, sensitivity and robustness analysis to assess model assumptions, parameter recovery and performance under alternative data-generating processes.
- Built version-controlled research workflows with R, MATLAB, Git/GitHub and executed large-scale computational jobs on Linux HPC infrastructure.
- Documented model specifications, assumptions, estimation procedures, validation results and limitations in technical reports for review by collaborators and conference audiences.

Optimal Probabilistic Forecasts for Risk Management 2020 - 2022

- Developed and validated tail-focused predictive models across 12 S&P equity indices, using out-of-sample benchmarking to assess Value-at-Risk (VaR) and Expected Shortfall accuracy.
- Improved conditional VaR coverage substantially at the 2.5% and 5% risk levels and applied tail-calibrated VIX forecasts to a dynamic VIX-futures hedging strategy, increasing its Sharpe ratio from 0.206 to 0.318.

Option-Implied Downside and Upside Market Risk Measures 2024 - 2025

- Led the project in collaboration with external academics, overseeing model development, validation, simulation design and empirical analysis.
- Developed forward-looking asymmetric systematic risk measures from option prices using option pricing theory and numerical optimisation techniques to investigate the pricing of downside and upside systematic risk in the cross-section of equity returns.
- Processed and analysed large-scale equity return and OptionMetrics option datasets in R and MATLAB, and designed simulation studies to evaluate model performance and robustness under varying market conditions.

WORK EXPERIENCE

Teaching Associate

Monash Business School, Feb 2023 - Present

- Teach applied SQL, econometrics and statistical learning, covering relational-data analysis, data validation, clustering, cross-validation, time-series prediction, model selection and dimensionality reduction.
- Coordinated tutorial teams and unit delivery as a head tutor across multiple units.

Research Assistant (Modeller)

Monash Business School, Aug 2023 - Oct 2024

- Conducted economic and financial market analysis for the Thai Bond Market Association, using high-frequency bond market data to evaluate the impact of regulatory and policy changes.
- Developed linear and logistic time-series models with bootstrap inference, hypothesis testing and robustness analysis to evaluate bond-price jumps around policy changes and communicate decision-oriented insights.
- Developed reproducible R workflows to clean, align and validate imperfect bond price data, addressing missing observations, irregular timestamps and anomalous values.

Research Assistant (Data Analyst)

Melbourne Business School, Aug 2022 - Jan 2023

- Built a Python pipeline to retrieve and process high-frequency S&P 500 constituent data through the Refinitiv API, automating large-scale dataset construction and improving reproducibility.
- Developed reusable scripts in R for millisecond-level equity data and implemented quality checks for missing observations, timestamps, duplicate records and anomalous returns.
- Collaborated with academic stakeholders to translate project requirements into structured datasets and deliverables.

Macroeconomics Research Analyst Internship

Liaowang Think Tank, July 2021 - Oct 2021

- Conducted macroeconomic research on fiscal and monetary policy, evaluating their implications for economic activity.
- Produced written research reports and presentations, communicating economic insights to non-technical stakeholders.

EDUCATION

2023 - 2026	PhD in Financial Econometrics - Monash University
2021	Graduate Certificate in Economic Analysis - The University of Sydney
2017 - 2020	Bachelor of Commerce (Honours) in Finance and Econometrics - Monash University
	<i>Relevant coursework:</i> Fixed Income Analysis, including corporate and non-retail credit analysis, probability of default, loss given default, expected loss, and structural and reduced-form credit-risk models.

SELECTED HONOURS & AWARDS

2025	Prestigious International Conference Award, Monash Business School
2025	CFM/SoFiE Scholarship, Capital Fund Management & Society for Financial Econometrics
2024	Best Poster Runner-Up, Time Series and Forecasting Symposium , the University of Sydney
2023	Co-funded Monash Graduate Scholarship, Monash University